# V-Track Pro

Fleet, Finance and Operations Platform
for Electric Bikes and Tricycles in Ghana

Project Idea and Scope

Version 1.0 | 7 November 2026

Prepared for the Project Sponsor, Mr. Kwasi Mpiani-Brobbe

# 1. Executive Summary

This document sets out the scope of the system requested by the Project Sponsor, based on the project call of 18 October 2026 and subsequent correspondence. The working name is V-Track Pro, pending a final name. It is a locally owned software platform to track, control and manage a fleet of electric bikes and tricycles in Ghana, and to run the business around them: container sales, leases, Work-and-Pay collections, investor reporting, hub operations and battery charging.

It responds to the Sponsor's question of 17 August 2026: can a locally built tracking and fleet system for green transport replace foreign software with recurring fees in foreign currency? The answer is yes. The tracking engine has already been built. The remaining work is the business layer described in the Tier 1 and Tier 2 model.

### Key facts

- **Project Sponsor:** Mr. Kwasi Mpiani-Brobbe. Objective: local ownership of the technology that runs green transport.
- **Lead developer:** Samuel Appiah Kubi, working remotely.
- **Timeline:** the test run was first targeted for the end of September 2026. That date has passed; the MVP is now targeted for the end of November 2026.
- **Pilot:** three hubs first, in Greater Accra, Eastern, Central and Ashanti regions, growing to all 16 regions.
- **First imports:** the first containers of Huodi e-bikes and YJJ e-tricycles were expected about 60 days after 29 August 2026.

# 2. Guiding Principles

| Principle | What it means for the system |
|---|---|
| **Local ownership** | Self-hosted in Ghana; no dependence on foreign SaaS for tracking or data. Third parties only where unavoidable (mobile money, SMS, tracker hardware). |
| **Robust and secure** | Strict roles, encrypted device traffic, audited commands, rate limiting. Designed to resist abuse from inside and outside. |
| **Usable everywhere** | Works across all 16 regions, including areas with poor connectivity; simple screens for hub managers and riders. |
| **Data as an asset** | Collect and keep sales, payment, route and battery data, and plan for storage and its location. |
| **Safety first** | Never cut power to a moving vehicle; every enforcement action is traceable. |

# 3. The Business Model

The business operates in two tiers. The platform must serve both.

### Tier 1: Import and sales

- On demand, a fundraiser imports a 40 ft container from a Chinese supplier. It may be a mix of the four vehicle types or a single type.
- After clearance at Tema, the company sells to individuals and SMEs through **outright purchase**, **lease** or **Work-and-Pay**.
- **Requirement:** the system holds all sales and payment data for each consignment.

### Tier 2: Operations

- Investors (individuals or SMEs) own vehicles under outright purchase, lease or Work-and-Pay; riders are assigned to the vehicles.

- The four vehicle types differ in how revenue is collected, in payback figures and in payback periods.
- Charging is central. At first it uses ECG grid power, which remains cheaper than daily fuel. Solar-powered central charging points follow.
- Hubs are set up per region, town or assembly, each run by an assigned Hub Manager. Three hubs are used for the pilot.

### Vehicle types in scope

| Type | Brand or examples | Use |
|---|---|---|
| Electric bike | Huodi | Deliveries and passenger transport |
| Electric tricycle (passenger) | Pragyaa, Bajaj, Apsonic, YJJ | Passenger transport |
| Electric tricycle (goods) | Aboboyaa, YJJ | Goods transport |
| Electric tricycle (waste) | To be confirmed with supplier | Garbage collection, including gated communities and narrow alleys |

## 4. What the System Must Hold and Report

The Sponsor specified the data the system must provide:

| Area | Required data |
|---|---|
| **Consignments** | Every container: contents, units, sales per unit, payment details |
| **Riders** | Rider information, assigned vehicle, daily route details |
| **Charging** | Daily battery charge payments |
| **Revenue** | Daily sales deducted at source, or accessible to the payment platform or wallet |
| **Rider earnings** | Daily chop money plus any extra |
| **Summaries** | Weekly and monthly summaries per rider, vehicle, hub and investor |
| **Tracking** | Location, ID, battery level and status for every vehicle |

### Garbage collection mode

A new design is required to handle requests from gated communities, building on what the platform already has. Initial concepts: a request and scheduling flow for estates, a route and pickup log per vehicle, proof of collection, and a view for estate managers. A short discovery session with the Sponsor is needed before this module is specified.

## 5. Solution Overview

Five modules make up the platform. The first is largely built; the rest are the work for the MVP and afterwards.

| Module | Purpose | Status |
|---|---|---|
| **A. Tracking and security** | Live map, geofences, alerts, commands, immobilizer safety, rider ID, battery events | Built; needs hardware integration |
| **B. Sales and consignments** | Containers, units, buyers, outright, lease and Work-and-Pay contracts, payment records | Not built |
| **C. Hubs and operations** | Hubs, Hub Managers, rider assignment, daily routes, charging events and payments | Not built |

| Module | Purpose | Status |
| --- | --- | --- |
| **D. Finance and reporting** | Wallet or payment-platform link, daily deductions, chop money, investor payback, weekly and monthly summaries | Partly built (daily debit with mock gateway) |
| **E. Waste collection** | Estate requests, scheduling, route and pickup proof | Not built; needs design |

# 6. Technical Architecture

The existing platform is already modular and multi-tenant, so the new business modules plug in beside it.

| Layer | Technology |
|---|---|
| **Backend API** | NestJS with TypeScript and TypeORM; JWT authentication with refresh rotation and 2FA; role-based access |
| **Operational database** | PostgreSQL with PostGIS |
| **Position history** | TimescaleDB, kept separate from operational data |
| **Live state and queues** | Redis (digital twin per vehicle) and BullMQ (scheduled and delayed jobs) |
| **Device messaging** | TCP listeners for GT06, TK103 and Teltonika; MQTT; SMS wake-up fallback |
| **Real-time** | Socket.IO with per-organization rooms; a reconnecting client built for weak networks |
| **Frontend** | Static HTML admin app with a top navigation bar, plus a public site; no build step |
| **Intelligence** | Anomaly detection service (design), for example for battery tampering |

# 7. Hardware and Battery Integration

The supplier's details change how locking works, so this is the most important technical question.

| Fact from the supplier | Implication |
|---|---|
| 72V 100Ah lithium battery, removable and swappable (raise the hydraulic rear bin to take it out) | Battery theft and swapping must be tracked per rider and per charge |
| 220V dedicated smart charger | Charging events and payments are tied to hubs and charge points |
| Only lithium batteries have the anti-theft function | All vehicle versions will use lithium |
| Battery has a built-in GPS module; leaving a set safe distance cuts power automatically | The immobilizer may be the battery itself, not a relay we control. We must learn its protocol and whether it takes commands |

**Required from the Sponsor:** the supplier's protocol document for the GPS module, its server or app, and confirmation of whether commands can be sent to a server hosted locally. If the module cannot report to the local server, a separate GT06-compatible tracker will be fitted in parallel.

# 8. Work-and-Pay and Safe Immobilization

Under Work-and-Pay, the rider is debited each day through mobile money (MTN MoMo, Telecel Cash, AirtelTigo Money) at 6 PM. The Sponsor's requirement is that the vehicle locks if the debit fails. This is implemented with a safety grace period rather than an instant lock, so that a vehicle that is moving or carrying passengers is never locked. The payment gateway is a mock today; the real aggregator (for example Hubtel or Paystack) must be confirmed before launch.

| Step | Behaviour |
|---|---|
| **1. Debit** | At 6 PM Africa/Accra the system requests a debit per active rider through a payment gateway abstraction |
| **2. Failure** | State changes to payment failed; the rider receives an SMS warning (WhatsApp later) |
| **3. Grace period** | Default 12 hours; a successful payment at any time clears the state |
| **4. Safety check** | At expiry the system reads the vehicle's speed from its live state. If it is moving above 5 km/h, it waits and retries every 5 minutes |

| Step | Behaviour |
| --- | --- |
| **5. Immobilize** | Only when the vehicle is stationary does the system send the command, which is logged and audited |

# 9. Data Model Additions

New records needed for modules B, C and D. Names are proposals.

| Entity | Key fields |
|---|---|
| **Consignment** | Container reference, supplier, arrival date, clearance status, cost |
| **Consignment item** | Vehicle type, quantity, unit cost, unit price, tracker or battery serials |
| **Sale contract** | Buyer (investor), vehicle, mode (outright, lease, Work-and-Pay), price, deposit, schedule |
| **Investor** | Individual or SME, contact details, owned vehicles |
| **Hub** | Region, town, assembly, Hub Manager, charge points, status |
| **Rider assignment** | Rider, vehicle, hub, start and end dates, daily target |
| **Charge event** | Battery, hub, kWh or fee, payer, time, payment status |
| **Daily settlement** | Rider, vehicle, date, collected, deducted, chop money, extra, balance |
| **Wallet or ledger entry** | Source, destination, amount, reference, status (immutable) |
| **Waste request** | Estate, schedule, vehicle, pickup proof |

Already in place: riders, rider sessions, payment logs, immobilizer state, vehicles, devices, geofences, alerts, commands, audit logs and tenant limits.

# 10. Security and Data Sovereignty

- **Hosting in Ghana:** a local data centre such as Rack Centre or MainOne, or a government cloud, so data stays in the country and supports Data Protection Commission compliance.
- **Access control:** roles for administrator, hub manager, investor, rider and auditor, with tenant isolation.
- **Device security:** authenticated devices, encrypted MQTT payloads, replay protection and rate limiting.
- **Financial integrity:** an append-only ledger, idempotent payment references and reconciliation against the aggregator.
- **Privacy:** riders' location and payment data are personal data; collect only what is needed and keep an audit trail of who viewed it.
- **Backups:** encrypted, tested restores, stored in-country.

# 11. Roadmap

About three weeks remain until the end-of-November MVP. The MVP scope is kept small enough to complete and prove in a pilot hub.

| Phase | Scope |
|---|---|
| **Phase 1: Pilot MVP (end November)** | Admin login and roles; vehicles and trackers; live map; consignments and sale contracts; investors and riders; hubs and assignments; daily settlement with chop money; weekly and monthly summaries; Work-and-Pay with warning and safe immobilization; deployment on Ghana hosting |
| **Phase 2: Pilot hardening** | Real mobile money aggregator; battery GPS module integration; charge payments; rider ID with NFC; WhatsApp dispatch |
| **Phase 3: Scale** | More hubs and regions; solar charge points; waste-collection module; investor portal; local-language messages |

# 12. Risks

| Risk | Mitigation |
|---|---|
| Short timeline and remote development | Strict Phase 1 scope; weekly review calls; shared task sheet |
| Supplier battery module cannot report to our server | Obtain the protocol now; fit a separate tracker as fallback |
| Wrongful lock of a vehicle that is in use | Grace period, speed gate, retry loop, audit trail, manual override by a manager |
| Unreliable mobile money or telco connections | Idempotent debits, retries, reconciliation, SMS fallback |
| Weak connectivity in regions | Offline buffering on devices, reconnecting clients, light pages |
| Scope growth (waste collection, investor portals) | Keep to the phase plan; design discovery first |
| Legal and registration gaps | Confirm requirements early (see section 13) |

# 13. Open Questions and Dependencies

### Open questions raised by the Sponsor

- **App name:** to be decided; a shortlist of distinctive options will be proposed.
- **Third-party vendors:** mobile money aggregator, SMS provider, tracker supplier, hosting provider.
- **Telecom services:** SMS and data SIMs for trackers; mobile money access.
- **Registrations:** to be confirmed with the Sponsor and a legal adviser, including data protection registration and any transport or payment rules. Specific requirements have not yet been verified.

### Information required from the Sponsor

- Which hubs are the three pilot hubs, and how many vehicles and riders in each?
- Which vehicle types go into the pilot, and are lease and Work-and-Pay both live from day one?
- How does the daily money move: rider wallet, direct mobile money, or cash collected by the Hub Manager?
- Who can see what: investors, hub managers, riders?
- Is there a rider exception policy (sickness, leave, breakdown) before a vehicle may be locked?
- What are the volumes (vehicles, riders, hubs) in the first six months?

### Next steps

- Sponsor reviews this document and confirms the Phase 1 scope.
- Shared task sheet completed, with the pilot hub list and data fields added by the Sponsor.
- Sponsor requests the battery GPS module protocol from the supplier.
- Development of modules B to D begins on top of the existing platform.